Valuation Sketchbook

11 classes · Valuation · Fall 2026 finals

Each class: the idea in one line, the sketch step by step, plain explanation, formulas, a worked example, common traps and where to read more.

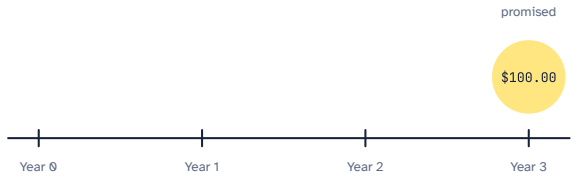
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Money today vs money later

A dollar later is worth less than a dollar today, because today's dollar could sit in a bank and grow. Present value (PV) just undoes that growth.

The sketch, step by step



Step 1. This is a timeline. Today is year 0. Someone promises you $100 at year 3.



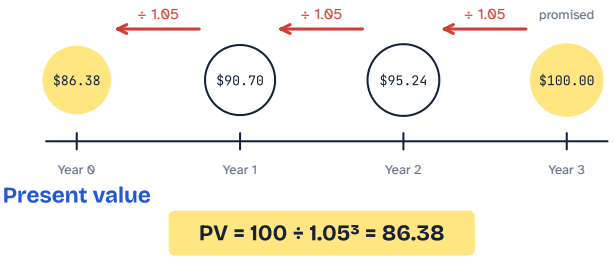
Step 2. Move the $100 back one year: divide by 1.05. It shrinks to $95.24.



Step 3. Back one more year: divide by 1.05 again. Now $90.70.



Step 4. Back to today: $86.38. That is the present value. Smaller coin, same "worth".



Step 5. Shortcut: do all three divides at once. PV = 100 ÷ 1.05^3 = 86.38.

What it means

Think of the bank as a time machine for money. Put $\$V_0$ in at 5% and next year you get $\$V_0 \times 1.05$. Present value runs the machine backwards: divide instead of multiply.

- **Future value (FV):** what money grows into. Multiply by (1 + r) each year.
- **Present value (PV):** what future money is worth today. Divide by (1 + r) each year.
- **Discount factor:** the number you multiply a future cash flow by to get its PV. For 3 years at 5% it's $1/1.05^3 = 0.8638$.
- **Annuity:** the same payment every period. **Perpetuity:** the same payment forever, PV = C / r.

- Prof. Shimko's point ("PV fails"): PV only works cleanly when you know the cash flows and the right rate. Risky cash flows need a risk adjustment, which later classes cover.

The formulas

```
PV = CF / (1 + r)^t
Perpetuity:  PV = C / r
Annuity:     PV = C × [1 - (1+r)^(-n)] / r
```

CF = cash flow, r = rate per period, t = number of periods, C = repeating payment.

Worked example

Question. You will get $100 in 3 years. The bank pays 5% a year. What is it worth today?

1. Write the formula: `PV = 100 / 1.05^3`
2. Work out $1.05^3 = 1.05 \times 1.05 \times 1.05 =$ `1.157625`
3. Divide: `100 / 1.157625 = 86.38`
4. Check by going forward: 86.38 × 1.157625 = 100. It works.

Answer: PV = $86.38

Calculator example

Inputs: Future cash flow ($) = 100, Rate per period (%) = 5, Periods = 3

```
Discount factor = 1 / 1.0500^3 = 0.8638
PV = 100.00 × 0.8638 = $86.38
```

Common traps

- Using an annual rate with monthly cash flows. Match the rate to the period (6% a year = 0.5% a month).
- Mixing up multiply (future) and divide (present).
- Forgetting that a perpetuity's first payment comes one period from now, not today.

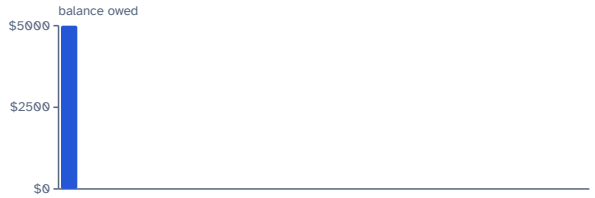
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_01_Lecture_1_Present_Value_Fails.pdf`
- `Valuation/02 Books and Readings/FRE6103_BOOK_03_Valuation_Principles_Ch_1_2.pdf`

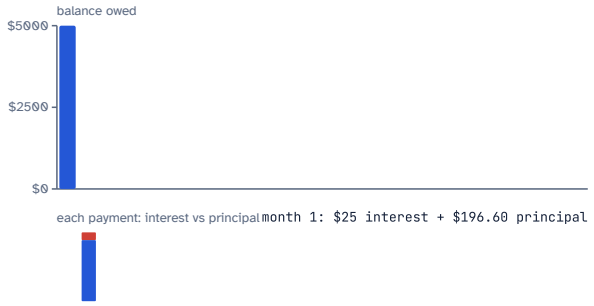
Loans: payments, default and prepayment

A loan payment is split into interest first, then the rest pays down what you owe. A pool of loans also loses money to people who default or pay off early.

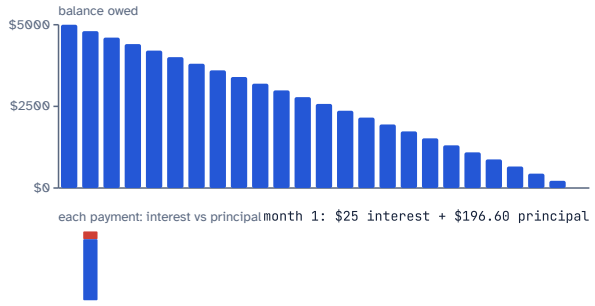
The sketch, step by step



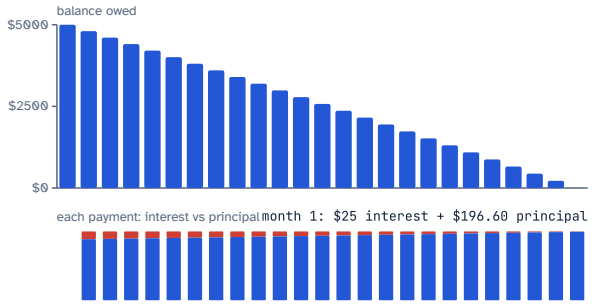
Step 1. You borrow $5,000. That tall bar is what you owe at month 0.



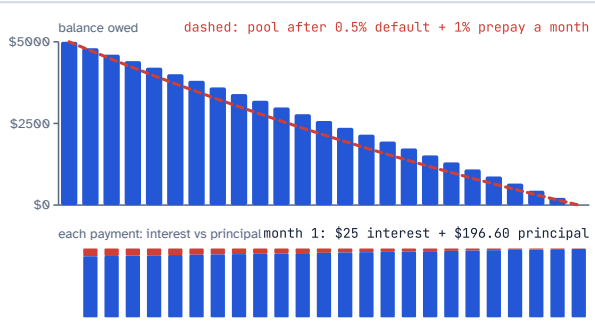
Step 2. Each month you pay $221.60. Red part = interest, blue part = paying down the loan.



Step 3. Over 24 months the balance falls to zero.



Step 4. Notice the red part shrinks and the blue part grows: less balance means less interest.



Step 5. For a lender with a pool of loans, defaults and prepayments make the pool (dashed) shrink faster.

What it means

This is the lecture's own example: borrow $5,000 for 24 months at 6% a year. Every month you pay the same amount. Early on most of it is interest, later most of it is principal, because the balance (and so the interest) keeps shrinking.

- **Amortization:** paying the loan down to zero with equal payments.
- **Borrower's view:** what does it cost me? **Lender's view:** what cash do I actually get back?
- **Default:** some borrowers stop paying. In the course model, a fixed share of the remaining pool defaults each month (0.5% in Mini Project 1).

- **Prepayment:** some borrowers pay everything early (1% a month in the project). The lender gets cash sooner but loses future interest.

The formulas

```
Payment:   PMT = B × i / [1 - (1+i)^(-n)]
Each month: interest = balance × i
            principal paid = PMT - interest
            new balance = balance - principal paid
Pool left after k months ≈ (1 - d - p)^k
```

B = amount borrowed, i = monthly rate (6%/12 = 0.5%), n = number of months, d = monthly default share, p = monthly prepay share.

Worked example

Question. Borrow $5,000 for 24 months at 6% a year. What is the payment, and what happens in month 1?

1. Monthly rate i = 6% / 12 = `0.005`
2. PMT = 5000 × 0.005 / (1 – 1.005^(–24)) = `$221.60` (Excel: PMT(0.06/12,24,–5000))
3. Month 1 interest = 5000 × 0.005 = `$25.00`
4. Principal paid = 221.60 – 25.00 = `$196.60`, new balance = `$4,803.40`

Answer: Payment $221.60; balance after month 1 is $4,803.40

Calculator example

Inputs: Amount borrowed ($) = 5000, Annual rate (%) = 6, Months = 24

```
Monthly rate = 0.5000%
Payment = $221.60
Month 1: interest $25.00, principal $196.60, balance $4,803.40
Total paid = $5,318.47, so total interest = $318.47
```

Common traps

- Using 6% instead of 0.5% for the month.
- Forgetting interest is paid before principal.
- In the pool model, default and prepayment both shrink the balance, but only default loses money.

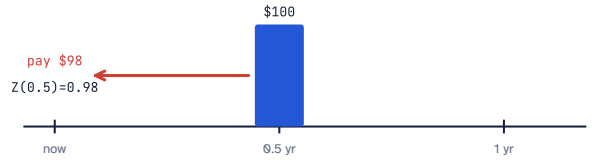
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_03_Lecture_2A_Loans_with_exponential_default_risk.pdf`
- `Valuation/07 Earlier Midterm Materials/FRE6103_REVIEW_02_Mini_Project_1_Two_Sides_of_a_Loan_Instructions.txt`

Bonds and building a zero curve

A zero-coupon bond pays one amount at one date, so its price tells you today's value of $1 at that date. Bootstrapping peels these discount factors out of coupon bonds one date at a time.

The sketch, step by step



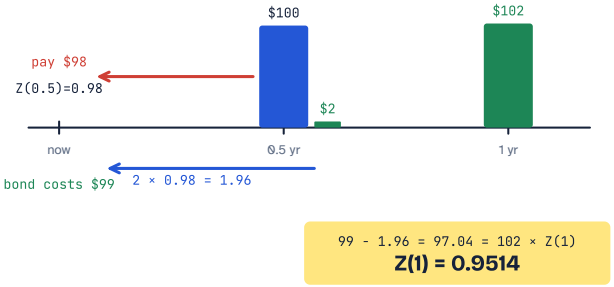
Step 1. Start with the easy one. A 6-month zero costs $98 for $100. So $1 in 6 months is worth $0.98 today.



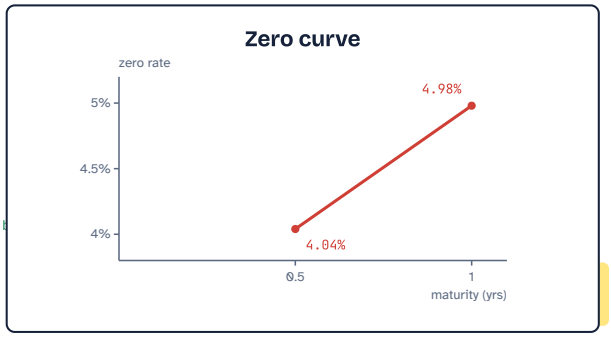
Step 2. Next, a 1-year coupon bond costing $99. It pays $2 at 6 months and $102 at 1 year.



Step 3. We already know how to value the $2: 2 × 0.98 = $1.96.



Step 4. The rest of the price must be the $102: 99 – 1.96 = 97.04. So Z(1) = 97.04 / 102 = 0.9514.



Step 5. Turn each Z into a rate and plot. That is your zero curve.

What it means

A coupon bond is just a bundle of little zero-coupon bonds, one for each payment. If you already know the discount factor for the early dates, you can subtract those pieces from the bond price. What's left must be the value of the last payment, which gives you the next discount factor. Repeat. That is **bootstrapping**.

- **Discount factor Z(t):** price today of $1 paid at time t.
- **Zero rate:** the rate that matches Z(t). Continuous: r = –ln Z(t) / t.
- **Zero curve / term structure:** zero rates plotted against maturity. Mini Project 2 builds this from Treasury data.

The formulas

```
Bond price = Σ CF(t) × Z(t)
Z(t) = e^(-r·t)      →   r = -ln Z(t) / t
Bootstrap last date:  Z(T) = (Price - Σ earlier CF × Z) / CF(T)
```

CF(t) = cash flow at time t, Z(t) = discount factor, r = zero rate (continuous).

Worked example

Question. A 6-month zero sells at $98 per $100. A 1-year bond with a 4% coupon paid twice a year ($2 every 6 months) sells at $99. Find Z(0.5) and Z(1).

1. Zero bond: `Z(0.5) = 98 / 100 = 0.98`
2. The 1-year bond pays $2 at 0.5 years and $102 at 1 year.
3. Value of the first coupon: `2 × 0.98 = 1.96`
4. What is left: `99 - 1.96 = 97.04 = 102 × Z(1)`, so `Z(1) = 0.9514`
5. Zero rates: −ln(0.98)/0.5 = `4.04%`, −ln(0.9514)/1 = `4.98%`

Answer: Z(0.5) = 0.98, Z(1) = 0.9514; zero rates 4.04% and 4.98%

Common traps

- Forgetting the final payment includes the coupon (102, not 100).
- Using the annual coupon (4) instead of the half-year coupon (2).
- Mixing continuous and annual zero rates.

Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_04_Lecture_2B_Bonds_and_rates.pdf`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_08_Treasury_data_090426.xlsx`
- `Derivatives/03 Lectures/FRE6073_LECTURE_05_Lecture_04_Interest_Rates.pptx`

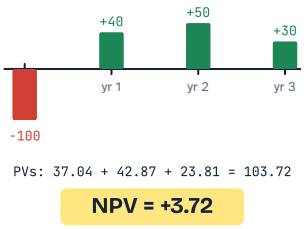
Capital budgeting: NPV and IRR

A company should take a project if its NPV is positive: the present value of what it brings in is more than what it costs. IRR is the discount rate where NPV is exactly zero.

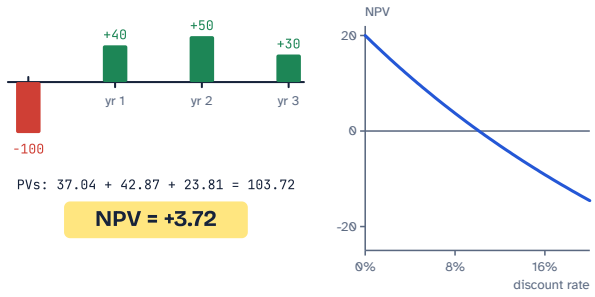
The sketch, step by step



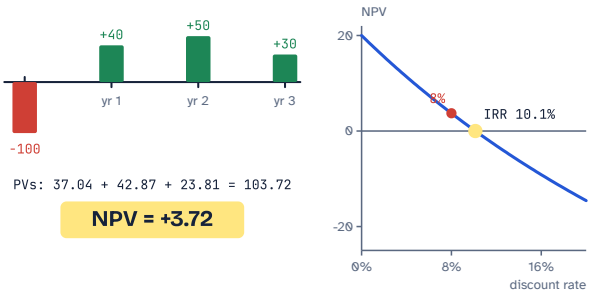
Step 1. Cash flows on a timeline: pay $100 now, get 40, 50, 30 later.



Step 2. Shrink each future amount to today's money at 8%, then add. NPV = +3.72.



Step 3. Now try every discount rate. NPV falls as the rate rises.



Step 4. Where the curve crosses zero (about 10.1%) is the IRR. Our 8% is to the left, so NPV > 0.

What it means

- **NPV:** add up the PV of every cash flow, including the (negative) cost today.
- **Discount rate:** the cost of capital, i.e. the return investors could get elsewhere at the same risk (CAPM from the Portfolio class helps here).
- **IRR:** the rate that makes NPV = 0. If IRR > cost of capital, NPV is positive. Find it by trial and error or Excel's IRR().
- **When IRR misleads:** projects of different size, cash flows that change sign more than once (can have two IRRs), or different timing. When in doubt, trust NPV.
- **Payback period** ignores timing and anything after the cutoff. Use it only as a rough check.
- Welch's book also covers **capital structure**: how mixing debt and equity changes the cost of capital (WACC).

The formulas

```
NPV = Σ CF_t / (1 + r)^t        (t = 0, 1, 2, …)
IRR: the r where NPV = 0
WACC = E/V × rE + D/V × rD × (1 - tax)
```

CF_0 is usually negative (the investment). E, D = market values of equity and debt, V = E + D.

Worked example

Question. A project costs $100 today and pays $40, $50 and $30 at the end of years 1, 2 and 3. Cost of capital 8%. NPV? IRR?

1. PV of year 1: 40 / 1.08 = `37.04`
2. PV of year 2: 50 / 1.08^2 = `42.87`
3. PV of year 3: 30 / 1.08^3 = `23.81`
4. NPV = 37.04 + 42.87 + 23.81 – 100 = `3.72` → take it
5. IRR (Excel IRR or trial): about `10.1%`, above 8%. Same answer.

Answer: NPV = $3.72, IRR ≈ 10.1%

Calculator example

Inputs: Cost today = 100, Year 1 = 40, Year 2 = 50, Year 3 = 30, Discount rate (%) = 8

```
NPV at 8% = 3.72 → take it
IRR ≈ 10.13%
```

Common traps

- Forgetting the year-0 cost.
- Choosing between two projects by IRR instead of NPV.
- Using a company-wide discount rate for a project with very different risk.

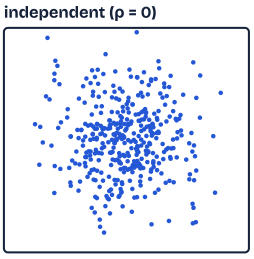
Where to read more

- `Valuation/02 Books and Readings/FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf`
- `Valuation/03 Lectures/FRE6103_LECTURE_13_VL10_Multiperiod_cash_flows.pdf`

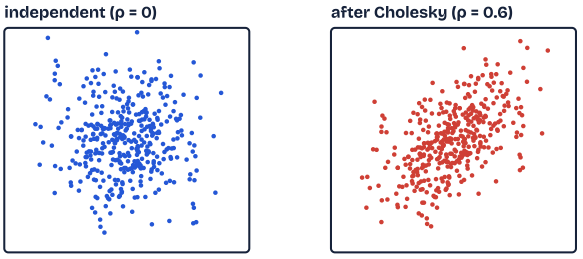
Simulation, correlation and Cholesky

Simulation means making up thousands of possible futures with random numbers and averaging the results. Cholesky is the trick that makes those random numbers move together.

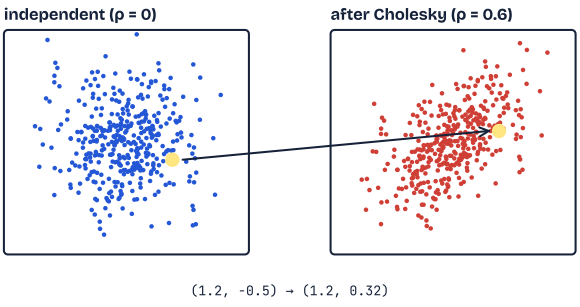
The sketch, step by step



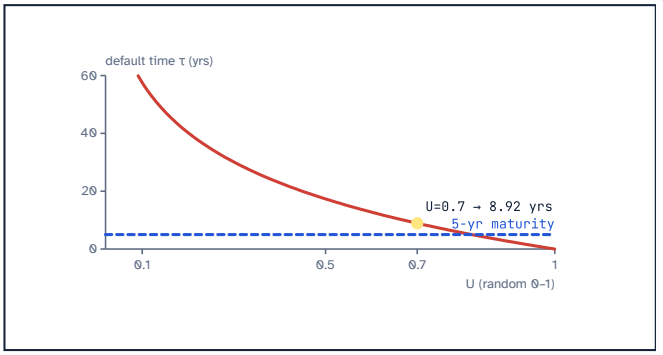
Step 1. 400 pairs of independent random numbers. A round cloud: knowing one tells you nothing about the other.



Step 2. Apply Cholesky with ρ = 0.6. The cloud stretches into a tilted oval. Now they move together.



Step 3. One dot: (Z1, Z2) = (1.2, −0.5) becomes (1.2, 0.32). It moved up because Z1 was high.



Step 4. Default times: draw U, then τ = −ln(U)/λ. Small U means a long wait; big U means early default.

What it means

Some problems are too messy for a formula (10 bonds defaulting at different times, a CDO waterfall). So you let the computer roll dice many times and look at the outcomes.

- **Uniform U** (between 0 and 1) → **normal Z** with NORM.S.INV(U) in Excel.
- **Correlation:** independent Z's don't move together. To get correlation ρ, mix them: X2 = ρ·Z1 + √(1−ρ²)·Z2. For many variables, the Cholesky matrix does this mixing for all of them at once.
- **Default time:** with a constant default rate λ, a random default time is τ = −ln(U)/λ. If τ is before maturity, that bond defaults in this run.
- **Sanity checks:** the sample mean, standard deviation and correlation should match what you asked for. "Moment matching" forces them to match exactly.

The formulas

```
Two variables:   X1 = Z1
                 X2 = ρ·Z1 + √(1 - ρ²)·Z2
Many variables:  X = L·Z   where  L·Lᵀ = correlation matrix
Default time:    τ = -ln(U) / λ
```

Z = independent standard normals, L = Cholesky (lower triangular) matrix, λ = default intensity per year.

Worked example

Question. ρ = 0.6, and you drew Z1 = 1.2, Z2 = −0.5. Find X1 and X2. Then: λ = 4% a year and U = 0.7. When does the bond default?

1. X1 = Z1 = `1.2`
2. √(1 − 0.36) = `0.8`
3. X2 = 0.6 × 1.2 + 0.8 × (−0.5) = 0.72 − 0.40 = `0.32`
4. τ = −ln(0.7)/0.04 = 0.3567/0.04 = `8.92 years`. For a 5-year bond: no default in this run.

Answer: X1 = 1.2, X2 = 0.32; default at 8.92 years

Common traps

- Using a covariance matrix when the formula needs a correlation matrix (or the other way round).
- Re-drawing random numbers every time Excel recalculates. Fix them (paste values) so results are comparable.
- Forgetting to check that sample correlation is near the target.

Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_05_Lecture_4_Simulation.pdf`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_02_Lecture_4_Cholesky_five.xlsx`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_03_Lecture_4_Simulation_spreadsheets.xlsx`

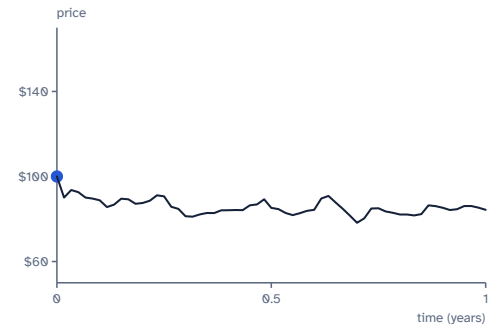
Random walks and stock price paths

A stock price is modelled as a steady drift plus random shocks. Using percentage shocks (geometric Brownian motion) keeps prices positive and gives a lognormal shape.

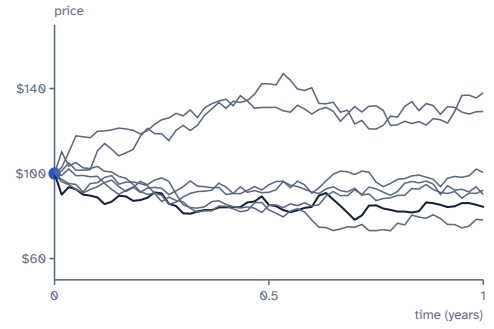
The sketch, step by step



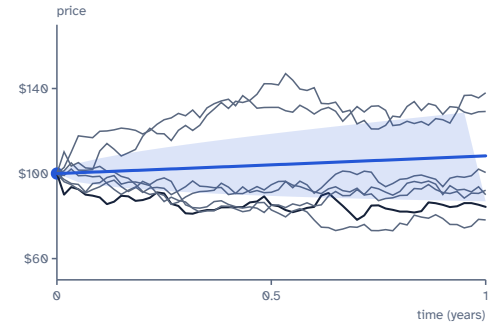
Step 1. Start at $100. Time runs left to right over one year.



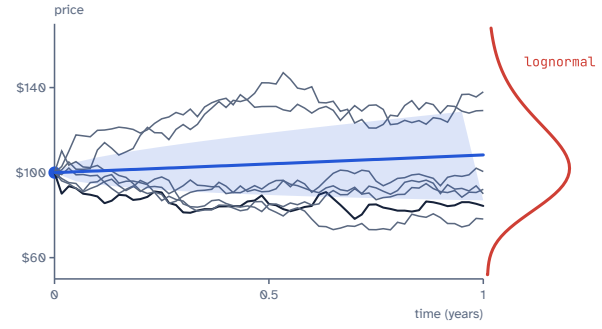
Step 2. One simulated path: small random up and down steps with a slight upward drift.



Step 3. More paths. Each one is a possible future.



Step 4. The blue line is the average path, 100·e^(0.08t). The band shows the typical spread, which widens over time.



Step 5. At year end the prices pile up in a lopsided (lognormal) shape: a long tail upward, never below zero.

What it means

- **Wiener process (Brownian motion) W:** a random walk where each tiny step is normal with variance equal to the time step. Over time Δt the change is ε√Δt.
- **Arithmetic Brownian motion:** $dX = \mu\,dt + \sigma\,dW$. Can go negative. Fine for spreads or rates, not for stock prices.
- **Geometric Brownian motion (GBM):** $dS = \mu S\,dt + \sigma S\,dW$. Returns are normal, prices are lognormal.
- **Simulating one step exactly:** $S_{next} = S \times \exp((\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}\cdot Z)$. The $-\sigma^2/2$ is there because of Itô's lemma; without it your average would be too high.
- **Mean reversion** (e.g. rates, commodities): the drift pulls the value back toward a long-run level.

The formulas

```
GBM:          dS = μ·S·dt + σ·S·dW
One step:     S(t+Δt) = S(t) × exp[(μ - σ²/2)·Δt + σ·√Δt·Z]
Expected:     E[S_T] = S0 × e^(μT)
```

μ = expected return (drift), σ = volatility, Z = standard normal draw.

Worked example

Question. S = 100, μ = 8%, σ = 20%, Δt = 1 year, and you draw Z = 0.5. What is the simulated price?

1. Drift part: (0.08 − 0.02) × 1 = `0.06`
2. Shock part: 0.20 × √1 × 0.5 = `0.10`
3. Add: 0.16, then `e^0.16 = 1.1735`
4. S = 100 × 1.1735 = `117.35`

Answer: S = $117.35

Common traps

- Forgetting $-\sigma^2/2$ in the exponent.
- Using σ × Δt instead of σ × √Δt.
- Mixing up the median (S0·e^((μ−σ²/2)T)) and the mean (S0·e^(μT)).

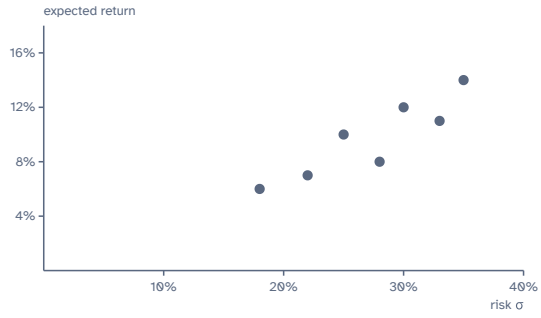
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_06_Lecture_5_Stochastic_Processes.pdf`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_05_Lecture_5_Simulation_spreadsheets.xlsx`
- `Hull 8e, Chapter 13 (Wiener processes and Itô)`

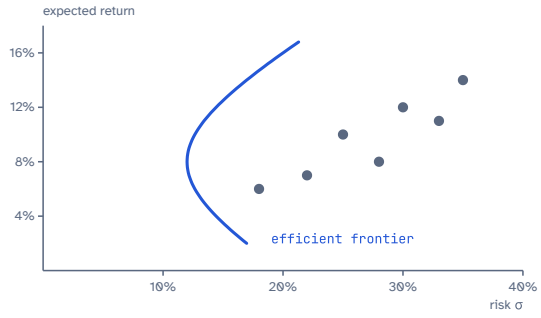
Portfolio theory and the CAPM

Investors only get paid for risk they cannot diversify away. Beta measures that risk, and the CAPM turns beta into a required return.

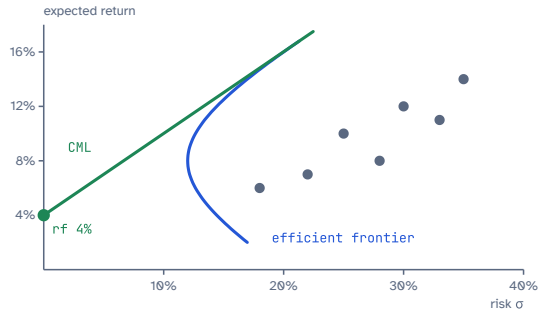
The sketch, step by step



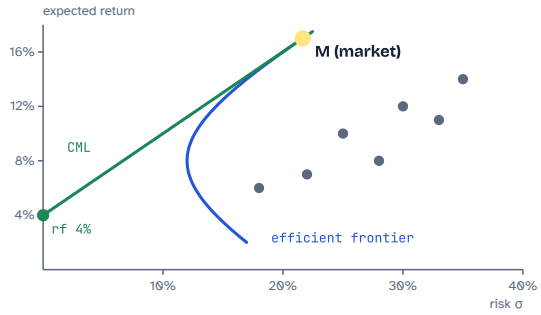
Step 1. Each dot is one stock: its risk (σ) and expected return.



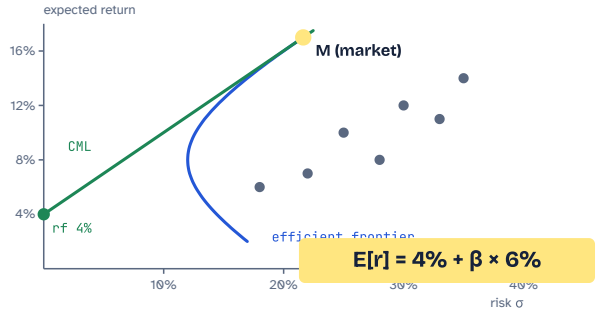
Step 2. Mixing them gives the efficient frontier: the best return for each level of risk.



Step 3. Add the risk-free asset at 4%. Draw a line from it that just touches the frontier.



Step 4. The touching point is the market portfolio M. Everyone should hold a mix of M and the risk-free asset.



Step 5. For one asset, CAPM says: required return = 4% + β × 6%.

What it means

- **Efficient frontier:** the best possible return for each level of risk, built by mixing stocks (Markowitz). Correlation below 1 bends it to the left.
- **Capital market line:** mix the risk-free asset with the best risky portfolio (the "market" M). It is the line from r_f that just touches the frontier. Its slope is the Sharpe ratio.
- **Beta:** how much an asset moves when the market moves. $\beta = \text{Cov}(r_i, r_M)/\text{Var}(r_M)$.
- **CAPM:** $E[r] = r_f + \beta \times (E[r_M] - r_f)$.

- **Lintner (certainty-equivalent) form:** value a risky cash flow X by subtracting a risk charge, then discounting at the risk-free rate. This is how the course values CDO tranches.

The formulas

```
Sharpe ratio:  (E[r] - rf) / σ
CAPM:          E[r_i] = rf + β_i × (E[r_M] - rf)
Lintner:       PV = (E[X] - λ·Cov(X, r_M)) / (1 + rf)
               λ = (E[r_M] - rf) / σ_M²
```

λ = market price of risk, X = an uncertain cash flow in one period.

Worked example

Question. rf = 4%, market return 10%, σ_M = 20%. (a) Required return for β = 1.2? (b) A cash flow has E[X] = $110 and Cov(X, r_M) = 2. Value it with Lintner.

1. (a) 4% + 1.2 × (10% − 4%) = `11.2%`
2. (b) λ = 0.06 / 0.04 = `1.5`
3. Risk charge = 1.5 × 2 = `3`, certainty equivalent = 110 − 3 = 107
4. PV = 107 / 1.04 = `102.88`

Answer: Required return 11.2%; PV = $102.88

Common traps

- Using total risk σ instead of beta for required return.
- Discounting the certainty equivalent at a risky rate. It already includes the risk charge, so use rf.
- Forgetting that correlation, not volatility alone, drives diversification.

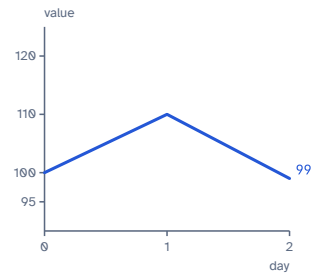
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_07_Lecture_6_MPT_and_the_CAPM.pdf`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_06_MPT_Ten_Stock_Portfolio.xlsx`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_07_sharpe_and_lintner.xlsx`

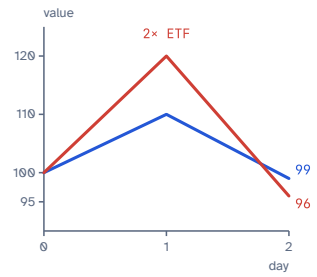
ETFs and futures for end users

Futures and ETFs look simple, but an end user faces hidden costs: rolling futures in contango, daily-reset leverage in leveraged ETFs, margin calls and basis risk.

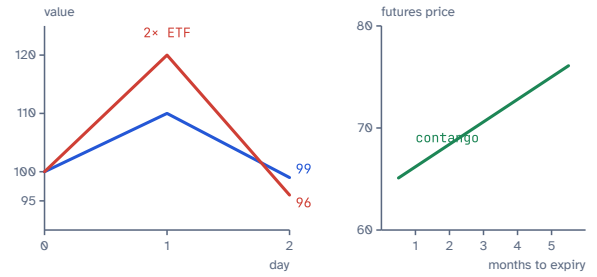
The sketch, step by step



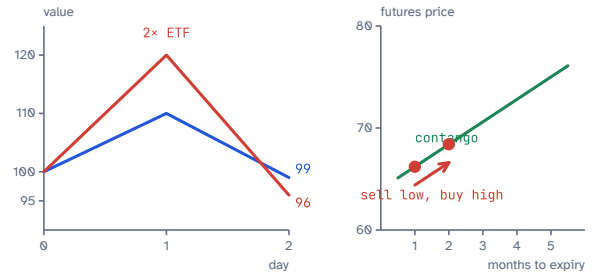
Step 1. An index that goes +10% then −10%: it ends at 99.



Step 2. A 2× daily ETF: +20% then −20%. It ends at 96, worse than "2 × −1%".



Step 3. Contango: futures prices rise with maturity. Each month the fund sells cheap near futures and buys pricier ones.



Step 4. That gap is lost every roll: a slow leak called negative roll yield.

What it means

- **ETF:** a fund that trades like a stock and tracks an index. Low cost, but commodity ETFs usually hold futures, not the commodity.
- **Roll:** futures expire, so a fund sells the near contract and buys the next one every month.
- **Contango** (futures above spot, curve upward): each roll buys higher and the fund slowly loses (negative roll yield). **Backwardation** (curve downward) gives positive roll yield.
- **Leveraged ETFs** (2×, 3×) reset every day. In a choppy market they lose value even if the index ends flat (volatility decay).
- **End-user risks:** margin/liquidity calls, basis risk, volume (volumetric) risk when you do not know how much you will need to hedge.

The formulas

```
Leveraged ETF over 2 days:  (1 + L·r1)(1 + L·r2)   ≠   1 + L·(total index return)
Roll yield (approx):         (F_near - F_next) / F_near  per roll
```

L = leverage (2 for a 2× ETF), r = daily index return.

Worked example

Question. An index goes +10% then −10%. What happens to the index and to a 2× daily ETF?

1. Index: 1.10 × 0.90 = `0.99` → down 1%
2. 2× ETF: day 1 = +20%, day 2 = −20%
3. 1.20 × 0.80 = `0.96` → down 4%
4. The ETF lost 4%, not 2× the 1% (2%). That extra loss is volatility decay.

Answer: Index −1%, 2× ETF −4%

Common traps

- Expecting a 2× ETF to give 2× the return over months.
- Ignoring roll cost when comparing a commodity ETF to the spot price.
- Forgetting that futures hedges can create big cash margin calls even when the hedge works.

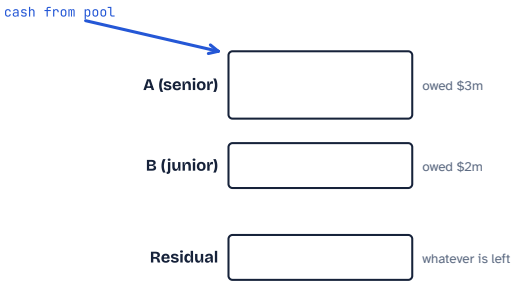
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_08_Lecture_7_ETFs_and_Futures_MARKUP.pdf`
- `Futures lecture recording (see "Watch online")`

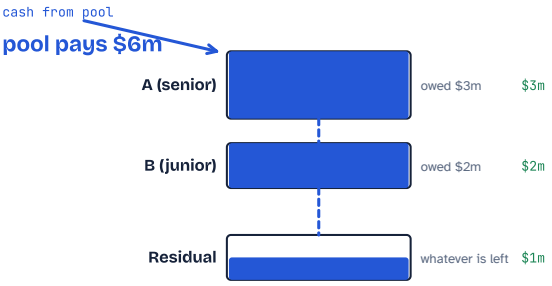
CDO waterfalls and tranches

A CDO pools many bonds and pays the cash out in order: senior tranche first, then junior, then whatever is left goes to equity. Who gets hurt depends on how many bonds default.

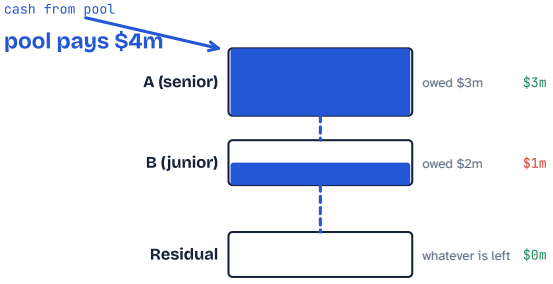
The sketch, step by step



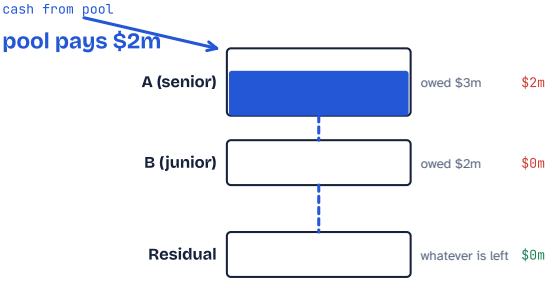
Step 1. Three buckets stacked like a fountain: A on top, B in the middle, residual at the bottom. Cash pours in from the pool.



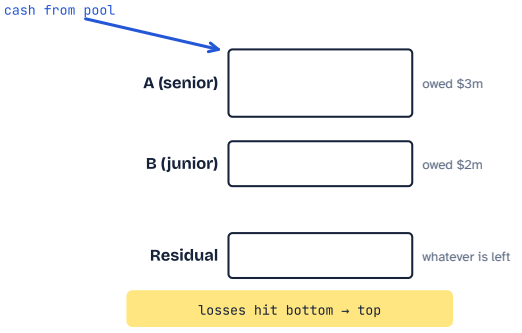
Step 2. Good period: the pool pays $6m. A fills ($3m), B fills ($2m), $1m spills into the residual.



Step 3. Some defaults: only $4m. A fills, B gets $1m of its $2m, residual gets nothing.



Step 4. Many defaults: $2m. Even A comes up short. B and residual get nothing.



Step 5. Rule: losses hit from the bottom up. Senior is protected by everyone below it.

What it means

- **Collateral:** the pool of bonds (Mini Project 3: ten $10m bonds, 6% coupons).
- **Waterfall:** each period, pay tranche A what it is owed, then B, then the residual. No carry-forwards in the project version.
- **Senior (A)** is safest; it only loses if many bonds default. **Equity/residual** takes the first losses.
- **Correlation matters:** with high correlation, bonds tend to default together. That makes "almost nothing" and "lots" of defaults both likelier. Senior tranches get *riskier*; equity can get *safer*.
- Value each tranche by simulating cash flows (Class 11) and pricing them with Lintner (Class 13).

The formulas

```
Pay A = min(cash, owed to A)
Pay B = min(cash - Pay A, owed to B)
Residual = cash - Pay A - Pay B
```

Repeat every period in every simulation run, then average and discount.

Worked example

Question. This period A is owed $3m and B is owed $2m. The pool pays (a) $6m, (b) $4m, (c) $2m. Who gets what?

1. (a) $6m: A gets 3, B gets 2, residual gets `1`
2. (b) $4m: A gets 3, B gets `1` (short by 1), residual `0`
3. (c) $2m: A gets `2` (short by 1), B `0`, residual `0`
4. So the residual is hit first, then B, and A only in bad cases.

Answer: (a) 3/2/1 (b) 3/1/0 (c) 2/0/0

Common traps

- Paying tranches pro rata instead of in order.
- Forgetting LGD: a defaulted bond still pays its recovery.
- Assuming correlation always makes every tranche riskier.

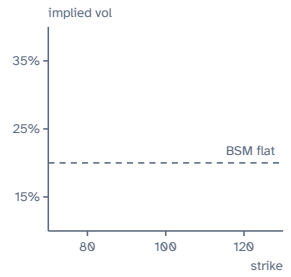
Where to read more

- `Valuation/07 Earlier Midterm Materials/FRE6103_REVIEW_04_Mini_Project_3_CDO_Risk_Part_1_Instructions.txt`
- `Valuation/06 Project and Recording Notes/FRE6103_NOTES_01_Project_and_recording_notes.md`
- `Valuation/03 Lectures/FRE6103_LECTURE_10_Lecture_03_Corporate_Bonds_and_CDOs.pdf`

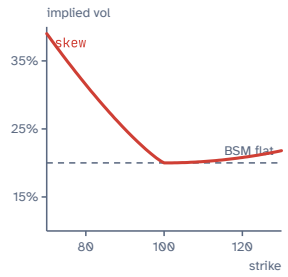
Volatility smiles and implied densities

If you back out σ from real option prices, it is not the same for every strike. That pattern tells you what the market really thinks future prices look like.

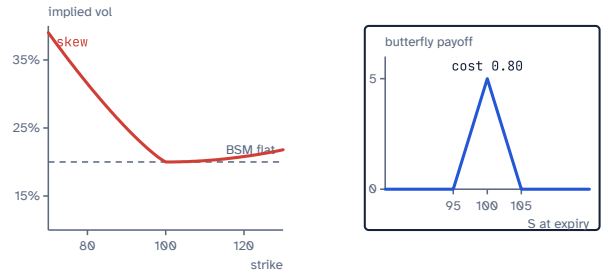
The sketch, step by step



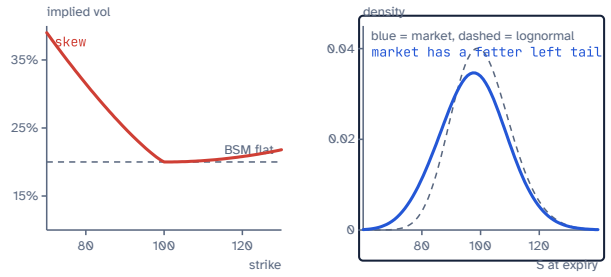
Step 1. Black-Scholes assumes one volatility for all strikes: a flat line.



Step 2. Real markets show a skew: low strikes (crash protection) have much higher implied vol.



Step 3. A butterfly (buy 95, sell two 100s, buy 105) pays a little tent. Its cost measures probability near 100.



Step 4. Line up many butterflies and you get the implied density. Compared with the lognormal (dashed), the left tail is fatter.

What it means

- **Implied volatility:** the σ you plug into Black-Scholes to match a market price. Solve it with Goal Seek.
- **Smile / skew:** plot implied vol against strike. For stocks, low strikes usually have higher vol (a skew): crashes are feared more than the lognormal model allows.
- **Volatility surface:** implied vol across strike *and* maturity.
- **Implied density (Breeden-Litzenberger):** the second derivative of call price with respect to strike gives the market's risk-neutral probability density. In practice: a butterfly spread.
- **European vs Asian:** an Asian option pays on the average price, which is less volatile, so it is cheaper. It usually needs simulation.

The formulas

```
Implied density:  q(K) = e^(rT) × ∂²C/∂K²
Butterfly approx: q(K) ≈ e^(rT) × [C(K-h) - 2·C(K) + C(K+h)] / h²
```

C(K) = call price at strike K, h = gap between strikes.

Worked example

Question. Calls cost 8.00, 5.00 and 2.80 at strikes 95, 100, 105. Estimate the probability density at 100 (ignore discounting). Roughly what is the chance S_T ends between 97.5 and 102.5?

1. Butterfly cost: 8.00 − 2 × 5.00 + 2.80 = `0.80`
2. Divide by $h^2 = 25$: density ≈ `0.032` per $1
3. Over a $5 window: 0.032 × 5 = `0.16`
4. So about a 16% risk-neutral chance of ending near $100.

Answer: Density ≈ 0.032 per $; about 16% chance

Common traps

- Reading a risk-neutral density as a real-world forecast.
- A negative butterfly cost means arbitrage or bad data, not a negative probability.
- Using one σ for every strike when the smile is steep.

Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_15_Volatility_Surface.pdf`
- `Valuation/03 Lectures/FRE6103_LECTURE_09_Lecture_9_BSM_and_Implied_Densities.pdf`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_01_European_and_Asian_option_benchmarking.xlsx`

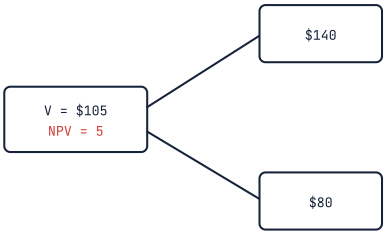
Real options and the Merton model

Choices in a business (wait, expand, abandon) are options, so they have value that plain NPV misses. Even a company's stock is an option on its assets.

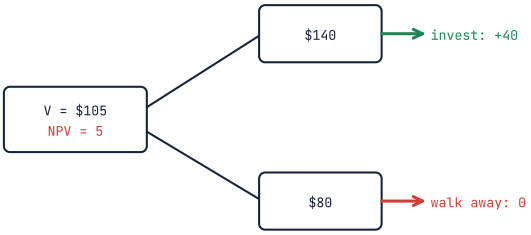
The sketch, step by step



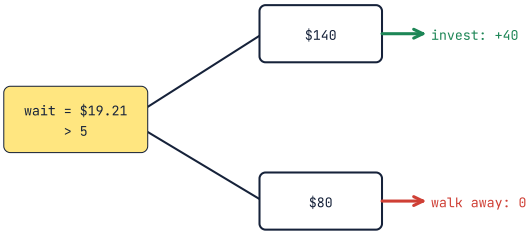
Step 1. Today the project is worth $105 and costs $100. NPV = $5.



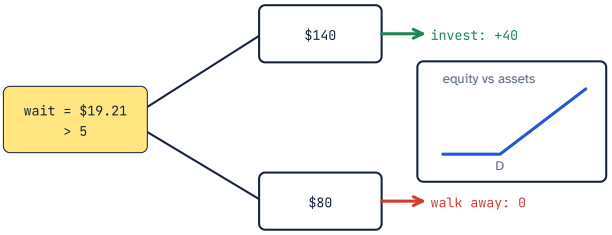
Step 2. Next year it is worth $140 (good) or $80 (bad).



Step 3. If you wait, you invest only in the good case: gain $40. In the bad case you walk away: $0.



Step 4. Rolled back with p = 0.504: the right to wait is worth $19.21. Waiting wins.



Step 5. Same idea for a firm: shareholders get max(Assets – Debt, 0). Equity is a call on the assets.

What it means

- **Real option:** the right, not the duty, to take a business action later. Waiting lets you invest only if things go well.
- Value it like Class 16: build a tree of the project value, find risk-neutral p, roll back.
- If the option value is bigger than investing now (NPV today), **wait**.
- **Energy example (VL14):** a power plant is a call on the spread between power and fuel prices: run it only when profitable.
- **Merton model:** equity = call option on the firm's assets with strike = debt. Default happens if assets end below the debt. **Distance to default** counts how many σ the firm is from that line.

The formulas

```
Risk-neutral p = [V0·(1 + rf) - V_down] / (V_up - V_down)
Option value   = [p·max(V_up - I, 0) + (1-p)·max(V_down - I, 0)] / (1 + rf)
Merton:  Equity = V·N(d1) - D·e^(-rT)·N(d2)
         DD = [ln(V/D) + (μ - σ²/2)T] / (σ√T),   PD = N(-DD)
```

V = value of project or firm assets, I = investment cost, D = face value of debt.

Worked example

Question. A project is worth $105 today and costs $100. Next year it will be worth $140 or $80; rf = 5%. Invest now, or keep the right to invest next year (cost still $100)?

1. NPV now = 105 − 100 = `5`
2. p = (105 × 1.05 − 80) / (140 − 80) = 30.25/60 = `0.504`
3. Wait: next year invest only if up: payoff 40 or 0
4. Option value = 0.504 × 40 / 1.05 = `19.21`. Bigger than 5, so wait.
5. Merton check: V = 100, D = 80, μ = 6%, σ = 25%, T = 1 → DD = (0.2231 + 0.0288)/0.25 = `1.01`, PD = N(−1.01) ≈ `15.7%`

Answer: Waiting is worth $19.21 vs $5 now. Merton PD ≈ 15.7%

Common traps

- Using the real-world probability in the roll-back.
- Comparing the option with zero instead of with investing now.
- In Merton, using the stock's volatility as the asset volatility.

Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_11_Options_in_a_valuation_context.pdf`
- `Valuation/03 Lectures/FRE6103_LECTURE_14_VL14_Real_Options_in_Energy.pdf`
- `Valuation/02 Books and Readings/FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf`
- `Risk syllabus RMN6 (Merton, distance to default)`

Watch and learn online

Free videos and courses to use alongside these classes. Coursera courses are free when you choose "audit".

Valuation

- **Your course: Week 1 recording (Prof. Shimko)** (Your own class video)
  https://www.youtube.com/watch?v=5t3T9z_fM4U
  Start here: it is your instructor explaining present value and the course setup.
- **Aswath Damodaran (NYU Stern): free valuation classes** (Free lecture videos, slides and spreadsheets)
  https://pages.stern.nyu.edu/~adamodar/New_Home_Page/home.htm
  His Valuation and Corporate Finance classes (also on his YouTube channel) cover DCF, discount rates, multiples and real options in plain language.
- **MIT OpenCourseWare 15.401: Finance Theory I (Andrew Lo)** (Free full lecture videos)
  https://ocw.mit.edu/courses/15-401-finance-theory-i-fall-2008/
  Present value, bonds, CAPM and capital budgeting: matches the Valuation classes almost one-to-one.
- **Yale (Coursera): Financial Markets with Robert Shiller** (Free to audit, video course)
  https://www.coursera.org/learn/financial-markets-global
  Good big-picture lectures on risk, portfolio theory, CAPM and how markets work.
- **Khan Academy: Finance and capital markets** (Free short videos, very beginner)
  https://www.khanacademy.org/economics-finance-domain/core-finance
  Units on interest and present value, bonds, CDOs and mortgage-backed securities.